
Phase 6:Project Testing

Credit Card Approval System

Testing Approach

Testing covered three areas: model evaluation on a held-out test set, form validation on the Flask front end, and manual end-to-end checks of the prediction flow from form submission through to result display.

Sample Test Cases

Test Case	Input	Expected Result
Valid low-risk applicant	Good credit history, stable income	Approved
Valid high-risk applicant	Past-due payment history	Rejected
Missing field	Income left blank	Form validation error
Boundary income value	Income at minimum threshold	Handled without crash
Unseen category	New occupation value	Falls back gracefully, no crash

Model Evaluation Results

Model	Accuracy	Precision	Recall	F1-score
Logistic Regression	0.762	0.71	0.69	0.70
Decision Tree	0.741	0.68	0.70	0.69
Random Forest	0.799	0.77	0.75	0.76
XGBoost (tuned)	0.811	0.79	0.78	0.78